

Machine Learning Assignment – Logistic Regression

1. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that allows computers to learn patterns from data and make decisions or predictions without being explicitly programmed.

Instead of writing fixed rules, machine learning models learn from examples and improve their performance over time.

Applications of Machine Learning

- Image Recognition
 - Spam Email Detection
 - Recommendation Systems
 - Voice Assistants
 - Self-driving Cars
 - Medical Diagnosis
-

2. Types of Machine Learning

2.1 Supervised Learning

In Supervised Learning, the model is trained using labeled data. The input and output are already known.

Examples:

- House Price Prediction
- Student Result Prediction
- Gender Classification

Algorithms:

- Linear Regression
 - Logistic Regression
 - Decision Tree
 - Random Forest
-

2.2 Unsupervised Learning

In Unsupervised Learning, the data does not contain labels. The model finds hidden patterns or groups.

Examples:

- Customer Segmentation
- Grouping Similar Products

Algorithms:

- K-Means Clustering
 - Hierarchical Clustering
-

2.3 Semi-Supervised Learning

Semi-Supervised Learning uses a small amount of labeled data and a large amount of unlabeled data.

Example:

- Medical image classification where only few images are labeled.

Advantage:

- Reduces labeling cost.
 - Improves model accuracy.
-

2.4 Reinforcement Learning

Reinforcement Learning is a learning method where an agent learns by interacting with the environment.

The agent receives:

- Rewards for correct actions
- Penalties for wrong actions

Examples:

- Game Playing AI
 - Robotics
 - Self-driving Cars
-

3. Logistic Regression

Introduction

Logistic Regression is a supervised machine learning algorithm used for classification problems.

It predicts categorical outputs such as:

- Yes / No
- True / False
- Male / Female

Although the name contains “Regression”, it is mainly used for classification.

Working of Logistic Regression

Logistic Regression uses the Sigmoid Function to convert values into probabilities between 0 and 1.

The sigmoid equation is:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Where:

- z = weighted sum of input features
- e = Euler’s constant

If probability $> 0.5 \rightarrow$ Class 1

If probability $< 0.5 \rightarrow$ Class 0

Advantages of Logistic Regression

- Simple and easy to implement
- Works well for binary classification
- Fast training process
- Easy to interpret

Disadvantages

- Not suitable for highly complex data
 - Sensitive to outliers
 - Assumes linear relationship
-

4. Logistic Regression using Gender Classification Dataset

Objective

To classify whether a person is Male or Female using Logistic Regression.

Python Program

```
# Import required libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Load dataset
# Replace the file name with your dataset file

data = pd.read_csv('gender_classification.csv')

# Display first 5 rows
print(data.head())

# Features and target
# Example columns: height, weight, shoe_size
X = data[['height', 'weight', 'shoe_size']]
y = data['gender']

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression()

# Train the model
model.fit(X_train, y_train)

# Predict using test data
y_pred = model.predict(X_test)

# Accuracy of model
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Classification Report
print(classification_report(y_test, y_pred))
```

Explanation of Each Step

1. Import Libraries

Used libraries for:

- Data handling
- Model training
- Accuracy evaluation

2. Load Dataset

The dataset is loaded using pandas.

3. Select Features and Target

Features are input values. Target is the output column.

4. Split Data

Dataset is divided into:

- Training data
- Testing data

5. Create Model

LogisticRegression() creates the ML model.

6. Train Model

fit() trains the model using training data.

7. Prediction

predict() is used to predict outputs.

8. Accuracy

accuracy_score() measures model performance.

5. Logistic Regression using Titanic Dataset

Objective

To predict whether a passenger survived or not using Logistic Regression.

Since preprocessing is already completed, the processed dataset is directly used.

Python Program

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Load preprocessed Titanic dataset

data = pd.read_csv('titanic_preprocessed.csv')

# Display first 5 rows
print(data.head())

# Features and target
# Survived column is target
X = data.drop('Survived', axis=1)
y = data['Survived']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression(max_iter=1000)

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Classification Report
print(classification_report(y_test, y_pred))
```

6. Conclusion

Machine Learning helps systems learn from data and make intelligent decisions.

In this assignment:

- Different types of Machine Learning were studied.
- Logistic Regression concepts were understood.
- Logistic Regression was implemented using:
 - Gender Classification Dataset
 - Titanic Dataset

The model successfully classified data and produced predictions with good accuracy.